CSE STUDY HUB

Data Structures & Algorithms

UNIT 3 · Linked lists and trees

What this unit is about

This unit builds understanding of **Linked lists and trees**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. Singly linked list: nodes, traversal, insertion and deletion

2. Doubly linked list and circular linked list; comparison and operations

3. Linked-list searching, updating and memory representation

4. Tree terminology: root, parent, child, leaf, degree, level and height

5. Tree traversals: preorder, inorder, postorder and level order

6. Binary Search Tree: search, insertion and deletion

7. Balanced trees: AVL rotations; B-Tree and B+ Tree concepts and uses

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 3 NOTES · PART 1

Singly linked list: nodes, traversal, insertion and deletion

A linked list stores data in nodes connected by pointers. It supports flexible insertion/deletion without shifting a whole array, but accessing the kth item requires traversal from the head (O(n)).

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Doubly linked list and circular linked list; comparison and operations

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Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Linked-list searching, updating and memory representation

Linked-list searching, updating and memory representation is a syllabus concept in Data Structures & Algorithms. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Linked lists and trees, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 3 NOTES · PART 2

Tree terminology: root, parent, child, leaf, degree, level and height

A tree is a hierarchical, connected acyclic structure. Learn root, leaf, parent, child, depth and height. Traversals visit nodes in a defined order; BST ordering enables efficient search when balanced.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Tree traversals: preorder, inorder, postorder and level order

A tree is a hierarchical, connected acyclic structure. Learn root, leaf, parent, child, depth and height. Traversals visit nodes in a defined order; BST ordering enables efficient search when balanced.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Linked lists and trees, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 3 NOTES · PART 3

Binary Search Tree: search, insertion and deletion

Binary search requires sorted data. Compare the target with the middle item and discard the half that cannot contain it. Repeating halves the search interval, giving O(log n) time; iterative search uses O(1) extra space.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Balanced trees: AVL rotations; B-Tree and B+ Tree concepts and uses

A tree is a hierarchical, connected acyclic structure. Learn root, leaf, parent, child, depth and height. Traversals visit nodes in a defined order; BST ordering enables efficient search when balanced.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Linked lists and trees, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **Singly linked list: nodes, traversal, insertion and deletion** — know the definition, core working and one example.

• **Doubly linked list and circular linked list; comparison and operations** — know the definition, core working and one example.

• **Linked-list searching, updating and memory representation** — know the definition, core working and one example.

• **Tree terminology: root, parent, child, leaf, degree, level and height** — know the definition, core working and one example.

• **Tree traversals: preorder, inorder, postorder and level order** — know the definition, core working and one example.

• **Binary Search Tree: search, insertion and deletion** — know the definition, core working and one example.

• **Balanced trees: AVL rotations; B-Tree and B+ Tree concepts and uses** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from Linked lists and trees. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define Singly linked list: nodes, traversal, insertion and deletion and explain its importance.

Q2. Explain the working/steps involved in Doubly linked list and circular linked list; comparison and operations.

Q3. Differentiate between two related concepts from Linked lists and trees.

Q4. Give one practical application of Balanced trees: AVL rotations; B-Tree and B+ Tree concepts and uses and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of Linked lists and trees.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.